

ICT 1209 Web Technologies

Mini Project: Interactive Web Application Development

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Rajarata University of Sri Lanka

Department of ICT — ICT 1209: Web Technologies

Mini Project — Phase 1: Proposal & Wireframes

Student Resource Hub

Project Title	EduNest-Student Resource Hub	
Course	ICT 1209 – Web Technologies	
Batch	2024 (First Year, Bachelor of ICT)	
Group Member 1	U.G.S.S. Udagedara	ITT/2024/108
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Submission Date	5th July 2026	
GitHub Repository	https://github.com/GeethakaBandara/EduNest-Student-Resource-Hub	

1. Purpose and Goals

The Student Resource Hub is a web-based platform that allows university students to upload, browse, search, and download academic resources such as lecture notes, past papers, and study guides in one centralised location. Many students currently rely on scattered WhatsApp groups, shared drives, and informal channels to exchange study materials, which makes resources hard to find, easy to lose, and difficult to verify for relevance. This project addresses that gap with a structured, searchable, and database-driven web application.

The main goals of the project are to:

- Provide a single, organised platform where students can upload and access course-related resources by faculty, subject, and category.
- Reduce the time students spend searching for reliable study material through fast, filterable search and browsing.
- Encourage peer-to-peer knowledge sharing through user accounts, resource ratings, and comments.
- Demonstrate a complete, full-stack, database-driven web application built with HTML5, CSS3, Bootstrap 5, JavaScript, PHP 8, and MySQL, in line with the ICT 1209 syllabus requirements.

2. Target Audience

The primary users are undergraduate students at Rajarata University of Sri Lanka, particularly first- and second-year students who are still building the habit of organised, independent study. Secondary users include:

- Senior students who wish to share notes, past papers, and study tips with juniors.
- Lecturers or teaching assistants who may point students towards curated resources for a specific module.
- Students from other faculties, since the category and tagging system is designed to scale beyond a single department.

3. Planned Features

The application will implement the following core features (minimum of three, as required by the guideline):

- **Resource Upload & Management** — Users can register, log in, and upload resources (PDF notes, slides, past papers) tagged with subject, category, and semester.
- **Search & Category Browsing** — A dedicated search and filter bar allows resources to be found instantly by keyword, subject, category, or file type using dynamic, client-side JavaScript filtering.
- **Ratings & Comments** — Registered users can leave comments and star ratings on uploaded resources, helping others identify the most useful materials.
- **User Authentication & Dashboard** — Secure sign-up and login with hashed passwords and session handling, giving each user a personal dashboard of their uploads and saved resources.
- **Contact / Feedback Form** — A simple contact form allows students to report issues, request resources, or reach the site administrators; submissions are stored in the database.

4. Technology Stack Confirmation

The project will strictly follow the required technology stack specified in the ICT 1209 Mini Project Guide:

Layer	Technologies
Frontend Structure	HTML5, CSS3, Bootstrap 5
Client-side Logic	JavaScript (Vanilla) — search/filter, form validation, image slider, smooth scrolling
Backend	PHP 8
Database	MySQL (via XAMPP)
Version Control	Git & GitHub

5. Planned Pages (Wireframes)

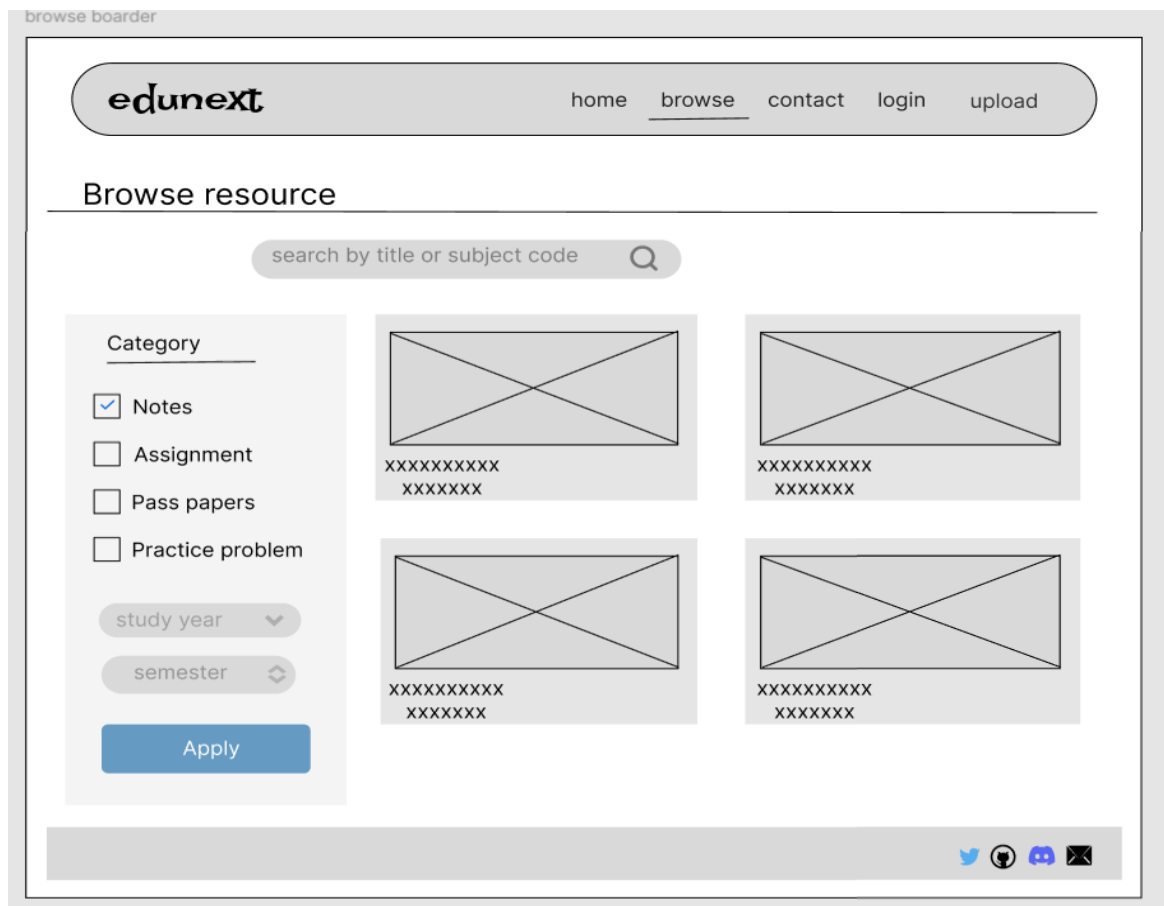
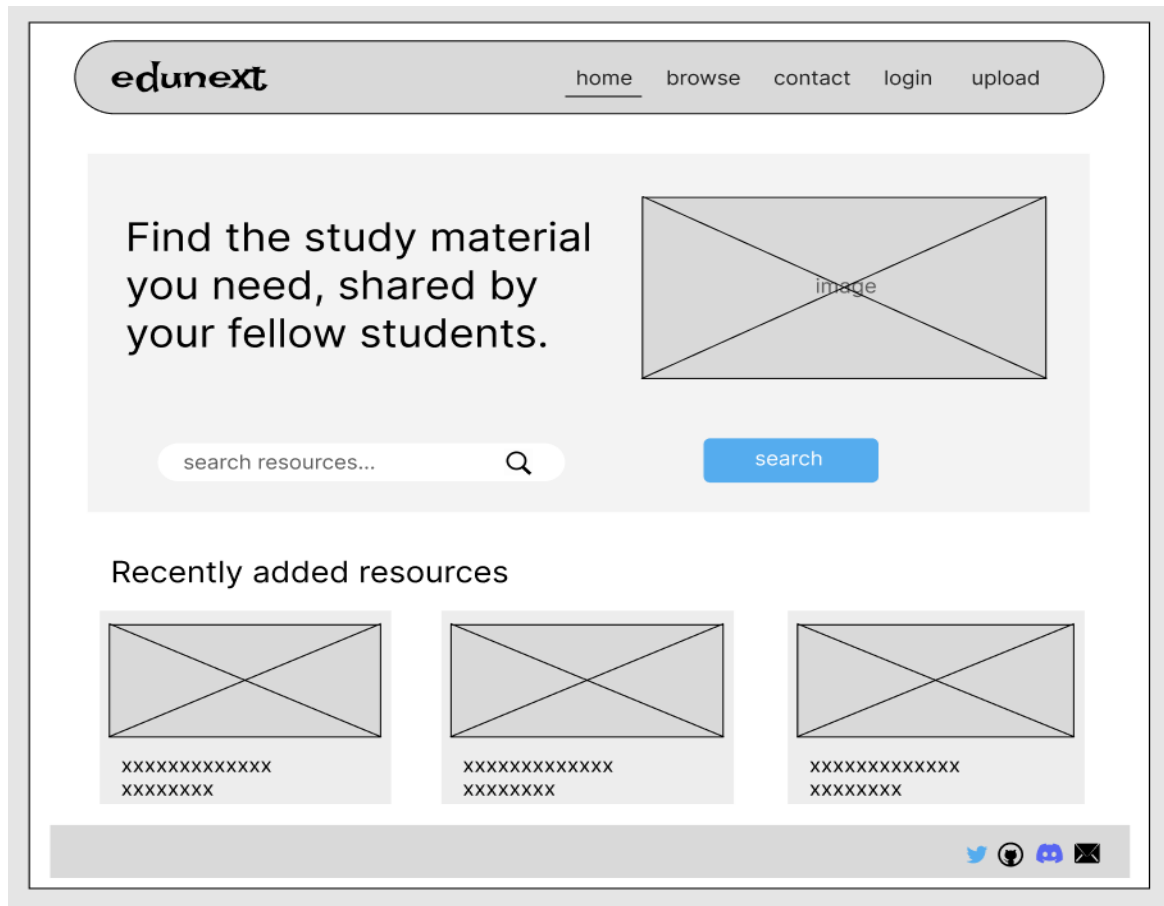
Wireframes have been prepared for the following core pages, attached as Appendix A of this submission:

- Home Page — hero banner, featured/recent resources, quick search bar, and navigation to categories.
- Browse / Resources Page — filterable and searchable grid/list of uploaded resources with category sidebar.
- Resource Detail Page — file preview/download, uploader info, comments, and star rating.
- Login / Register Page — authentication forms with client-side validation.
- Contact Page — feedback/query form with confirmation message on submission.

6. Conclusion

The Student Resource Hub aims to give Rajarata University students a reliable, centralised platform for finding and sharing academic materials, while giving our group a practical opportunity to apply the full ICT 1209 technology stack — from responsive front-end design through to secure PHP and MySQL backend integration. We look forward to developing this project across the three project phases and welcome any feedback on the proposed scope.

Wireframe Images



Contact

Thank you! your message has been sent successfully.

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Full name

Email address

Message

submit

